

ZENITH ? Project Documentation

Attack Path & Identity Privilege Graph Analyzer | Kurukshetra 2.0 (Domain 2)

1. Executive Summary & Problem Formulation

In modern enterprise Active Directory and Cloud IAM topologies, access permissions are rarely dangerous in isolation. However, chained together across intermediate identity compromises, credential caching in memory, and discretionary access control list (DACL) delegations, they form catastrophic multi-hop lateral attack paths to Domain Admins and root infrastructure.

Traditional compliance scanners evaluate permissions in silos and fail to detect indirect multi-hop paths. ZENITH formulates attack path discovery as an exact directed graph problem in Neo4j, using Dijkstra pathfinding and Cypher algorithms to compute deterministic exploit paths and attack costs.

2. Mathematical Model & Path Finding Complexity

ZENITH models identities as nodes V (Users, Groups, Computers) and privilege relationships as directed edges E with operational friction weights w in $[1, 10]$.

The primary path query evaluates candidate multi-hop walks between an initial identity and the objective using variable-length traversal MATCH $p = (\text{start})\text{--}[*1..6]\text{--}(\text{target})$, ranking paths by the summed exploit friction cost.

The algorithmic time complexity is $O(V * \log V + E)$, providing sub-15ms discovery across enterprise identity topologies.

3. Dynamic Risk Scoring Engine & Formula

ZENITH computes a real-time Risk Score in $[5, 100]$ using the formula: $\text{Score} = \text{clamp}(5, 100, \text{Base} - (\text{Cost} * 3.5) - (\text{Hops} * 2.5) + \text{SurfaceFactor})$.

Base Criticality: Tier-0 Domain Controllers and Admin Groups start at 100 baseline. Tier-1 infrastructure servers start at 90 baseline.

Deductions reflect operational friction: each hop increases detection probability (-2.5 pts), while exploit resistance subtracts 3.5 pts per friction weight unit.

Surface Factor: Active LSASS memory sessions and DACL ForceChangePassword rights add +10 pts exposure, whereas Credential Guard deducts 10 pts.

4. Decoupled AI Threat Narration Layer

ZENITH maintains a strict architectural separation: mathematical path discovery is NEVER delegated to probabilistic LLM inference. All paths, blast radii, and chokepoints are calculated deterministically by Neo4j graph algorithms.

The AI reasoning layer (NVIDIA NIM Llama-3.3-70b-instruct) operates strictly post-calculation to translate verified graph facts into actionable MITRE ATT&CK mappings (T1021.001 Remote Services, T1003.001 LSASS Memory, T1098 Account Manipulation) and CISO-ready remediation playbooks.

5. Blast Radius & Chokepoint Severance

Blast Radius Analysis computes all downstream reachable assets from any identity in the graph using BFS/DFS Cypher queries, categorizing reachable nodes by type and criticality tier.

Chokepoint Remediation Simulation enables SOC analysts to revoke candidate privilege edges (e.g. HasSession or CanResetPasswordOf) in real time to verify whether the attack path is eliminated or rerouted to a high-cost alternative.